



UNDER THE HOOD

Arpita Kapoor and Mohit Rangaraju
readersletters@thinkdigit.com

Most of you would wondered how Kratos (of God of War fame) is made so lively or how the whole backdrop of the game is made so involving. Although we wouldn't go to the extent to say that it's an easy job, we'll definitely say it's not rocket science either. With excellent open source software tools available, you can create your own game for free.

Don't get confused between a game and a game engine – these are two distinct entities. Consider a bike. It has its own engine. The same engine fitted into another body makes it a different bike altogether. Similarly, the game engine doesn't define your game. It just equips it with the necessary tools to feel like a game, such as game objects, characters, animation and artificial intelligence based codes.

Although game development is a multi-billion dollar industry, it's possible to create your own game using free and open source tools

Renderer and basic graphics pipeline flow

Rendering is the process of generating an image from a 3D model using a computer program. Simply put, without a renderer you won't be able to see anything at all. A renderer helps you visualise the scene and makes your game stand out.

How 3D models and worlds in your game are stored in memory is a function of the renderer. 3D objects are stored as points in a 3D world. These points called vertices are associated with each other in terms of surfaces and lines that make up triangles and polygons that recreate your 3D object. If you're rendering a cube using a game engine, it means you're drawing eight points that make up the vertices, twelve lines that make up the edges and six surfaces.

If, for example, you're making a 3D model of a car, you're dealing with thousands of polygonal surfaces. On the other hand, if you're talking of the